



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: QUANTUM COMPUTING – 24CS3207

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Quantum Computing for Hospital Appointment Scheduling	
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Abstract

Hospital appointment scheduling is a challenging task because hospitals need to manage many patients, doctors, departments, and available time slots. Traditional scheduling methods may result in appointment conflicts, long waiting times, and unused doctor slots. This project proposes a Quantum Computing-based Hospital Appointment Scheduling System to improve the scheduling process. The system uses a hybrid quantum-classical approach, where classical computing is used to collect and process patient and doctor information, while quantum computing is used for optimization. The main techniques explored in this project are Qubits, Quantum Superposition, Quantum Gates, Quantum Circuit, and Quantum Approximate Optimization Algorithm (QAOA) for finding suitable appointment schedules. The scheduling problem can be formulated as an optimization problem by considering doctor availability, patient requirements, appointment duration, and scheduling conflicts. The main goal is to reduce patient waiting time, avoid appointment clashes, and improve the utilization of doctors and hospital resources. This project demonstrates how quantum computing techniques can be applied to solve a real-world healthcare scheduling problem.